Methods for Structural Microeconometrics

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Welcome to this course! Find details on the syllabus here.

Here is a link to the git repo for this course, which by cloning is probably the simplest way for you to get access to the data, code, and assignments. I will continue to add materials throughout the course. You may prefer to selectively download these materials.

1 Topics

The course follows three parts of the online textbook:

1. Identification. Key readings: Keane (2010), Angrist and Pischke (2010), Marschak (1953), Heckman and Vytlacil (2007).
2. Discrete Choice, static and dynamic. Key readings: Rust (1987), Hotz and Miller (1993), Aguirregabiria and Mira (2002), Arcidiacono and Miller (2011).
3. Unobserved Heterogeneity. Key readings: Arcidiacono and Miller (2011), Bonhomme and Manresa (2015), Kasahara and Shimotsu (2009), Berry and Compiani (2023).

2 Assessment

Assessment is based solely on the class project, which is due at the end of the semester. The project asks you to pose a research question, write down a model that can answer it, argue that the model is identified, and demonstrate with Monte Carlo simulations that your estimator recovers the answer. See the project page for full details.

The project should be submitted as a single PDF by email.

2.1 Suggested Exercises

The exercises below mirror the problem sets from previous years of this course. They are for **self-guided learning only** and will not be assessed. Working through them is the best way to build the skills you will need for the project, and I am happy to discuss them in office hours. All of them can be found in the online textbook.

Topic	Exercise
Difference-in-differences and the labor supply elasticity	Exercise: Mr Straw Man
Identification of the labor supply model with instruments	Exercise: Labor Supply IV
Solving and estimating a dynamic discrete choice model	Exercise: Dynamic Labor Supply
Unobserved types and the EM algorithm	Exercise: EM with Types
Hidden Markov models and the forward-back algorithm	Exercise: HMM with Missing Outcomes
CCP estimation with welfare time limits	Exercise: CCP Estimation with Time Limits

3 Contact

For appointments and questions, please email.

Reading List

Aguirregabiria, Victor, and Pedro Mira. 2002. "Swapping the Nested Fixed Point Algorithm: A Class of Estimators for Discrete Markov Decision Models." *Econometrica* 70 (4): 1519–43.

Angrist, Joshua D., and Jörn-Steffen Pischke. 2010. "The Credibility Revolution in Empirical Economics: How Better Research Design Is Taking the Con Out of Econometrics." *Journal of Economic Perspectives* 24 (2).

Arcidiacono, Peter, and Robert A Miller. 2011. "Conditional Choice Probability Estimation of Dynamic Discrete Choice Models with Unobserved Heterogeneity." *Econometrica* 79 (6): 1823–67.

Berry, Steven T, and Giovanni Compiani. 2023. "An Instrumental Variable Approach to Dynamic Models." *The Review of Economic Studies* 90 (4): 1724–58.

Bonhomme, Stéphane, and Elena Manresa. 2015. "Grouped Patterns of Heterogeneity in Panel Data." *Econometrica* 83.

Heckman, James, and Edward Vytlacil. 2007. "Chapter 70 Econometric Evaluation of Social Programs, Part i: Causal Models, Structural Models and Econometric Policy Evaluation." In, edited by James J. Heckman and Edward E. Leamer, 6:4779–874. Handbook of Econometrics. Elsevier. https://doi.org/https://doi.org/10.1016/S1573-4412(07)06070-9.

Hotz, V Joseph, and Robert A Miller. 1993. "Conditional Choice Probabilities and the Estimation of Dynamic Models." *The Review of Economic Studies* 60 (3): 497–529.

Kasahara, Hiroyuki, and Katsumi Shimotsu. 2009. "Nonparametric Identification of Finite Mixture Models of Dynamic Discrete Choices." *Econometrica* 77 (1): 135–75.

Keane, Michael P. 2010. "A Structural Perspective on the Experimentalist School." *Journal of Economic Perspectives* 24 (2).

Marschak, Jacob. 1953. "Economic Measurements for Policy and Prediction." In *Studies in Econometric Method*, edited by W. Hood and C. Koopmans. John Wiley & Sons.

Rust, John. 1987. "Optimal Replacement of GMC Bus Engines: An Empirical Model of Harold Zurcher." *Econometrica: Journal of the Econometric Society*, 999–1033.